

# PIUM – Phase Interference Universe Model

## Low- $\ell$ CMB Anomalies and Phase Coherence

A Non-Tuned, Reproducible Empirical Test Framework Using Public Planck Data

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Year: 2026  
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### ABSTRACT

Several large-scale features have been persistently discussed in the low-multipole (low- $\ell$ ) regime of the cosmic microwave background (CMB), including quadrupole–octopole alignment, large-angle power suppression, non-trivial geometrical properties of certain multipoles, and localized non-Gaussian structures such as the Cold Spot (Planck Collaboration 2020 VII; Copi et al. 2010; Patel et al. 2025). Within the standard  $\Lambda$ CDM framework, these features are generally interpreted as statistical fluctuations, cosmic variance effects, or method-dependent anomalies.

This work does not introduce new dynamics, free parameters, or post hoc fitting procedures. Its objective is strictly methodological: to demonstrate that the low- $\ell$  CMB regime provides a suitable domain for a pre-defined, reproducible, non-tuned test of phase coherence using publicly available Planck data.

Within the Phase Interference Universe Model (PIUM), the large-scale CMB structure can be interpreted as a residual imprint of a globally organized phase field. However, this work does not assume that interpretation to be correct. Rather, it formulates a clear empirical question: whether the phase structure of low- $\ell$  modes is compatible with the random Gaussian hypothesis.

The primary contribution is the definition of a two-stage, non-tuned analysis protocol that allows this question to be tested in a controlled and falsifiable manner.

## 1. INTRODUCTION

In the standard cosmological model, the CMB is described, to first approximation, as a statistically isotropic, nearly Gaussian random field. Under this assumption, the phases of the spherical harmonic coefficients are independent and uniformly distributed (Planck Collaboration 2020 VII).

Over the past two decades, however, several large-scale features have repeatedly appeared in the literature. Among the most widely discussed are the alignment of the quadrupole ( $\ell = 2$ ) and octopole ( $\ell = 3$ ), the suppression of power at the largest angular scales, planarity and directional features of certain multipoles, and localized anomalies such as the Cold Spot (Copi et al. 2010; Patel et al. 2025; Aluri et al. 2025).

These observations are not generally regarded as decisive evidence for new physics. They are typically interpreted as statistical fluctuations or consequences of cosmic variance. Nevertheless, the persistence of these features across multiple data releases raises a narrower but well-defined question: whether the large-scale structure of the CMB is fully consistent with the simplest random-phase Gaussian picture.

## 2. PRECISE FORMULATION OF THE SCIENTIFIC QUESTION

The central question is:

Is the phase structure of low- $\ell$  CMB modes compatible with the random Gaussian hypothesis?

## 3. WHY THE LOW- $\ell$ REGIME?

The low- $\ell$  regime represents the largest observable angular scales in the CMB, where any global phase coherence would most naturally appear. At the same time, strong cosmic variance requires strictly pre-defined, non-tuned analysis.

## 4. OVERVIEW OF LOW- $\ell$ CMB FEATURES

Commonly discussed features include:

- quadrupole–octopole alignment
- large-angle power suppression
- planarity and directional anomalies
- entropy/tensor anomalies
- Cold Spot and related structures

### 4.1. SUMMARY TABLE OF LOW- $\ell$ CMB FEATURES

The following table summarizes commonly discussed large-scale CMB features in the low- $\ell$  regime. Values are indicative and provided for structural orientation.

| $\ell$   | D_ $_{\ell}$ ( $\mu K^2$ ) | Observed Feature  | Structural / Phase Note               |
|----------|----------------------------|-------------------|---------------------------------------|
| 2        | ~180–250                   | Low power         | Alignment with $\ell = 3$             |
| 3        | ~800–1100                  | Planar structure  | Shared preferred axis with $\ell = 2$ |
| 4        | ~1000–1200                 | Near average      | Weak correlations                     |
| 5        | ~1400–1500                 | Power excess      | Possible structural link              |
| 6–10     | ~900–1300                  | Mild anomalies    | Method-dependent deviations           |
| ~13,17   | –                          | Entropy anomalies | Possible anisotropy signals           |
| 20–22    | 600–800                    | Power dip         | Valley-like large-scale feature       |
| ColdSpot | –                          | Local minimum     | Non-Gaussian structure                |

References (by row):

|                |                                      |
|----------------|--------------------------------------|
| $\ell = 2-3$   | Planck 2020 V, VII; Copi et al. 2010 |
| $\ell = 3$     | Patel et al. 2025; Aluri et al. 2025 |
| $\ell = 4-5$   | Planck 2020 V                        |
| $\ell = 6-10$  | Billi et al. 2024                    |
| $\ell = 13,17$ | Patel et al. 2025                    |
| $\ell = 20-22$ | Planck 2020 V                        |
| Cold Spot      | Planck 2020 VII; Owusu et al. 2023   |

Notes

- Values are approximate (Planck 2018 maps)
- Cosmic variance is significant
- Features are not independent proofs

## 5. INTERPRETATIVE FRAMEWORK (PIUM)

PIUM interprets the universe as a globally organized phase field. The CMB may contain residual imprints of this structure. This is not assumed to be correct, only tested.

## 6. FORMAL FRAMEWORK

$$T(\theta, \phi) = \sum a_{\ell m} Y_{\ell m}$$

Null hypothesis: random Gaussian phases  
Alternative: non-random phase correlations

## 7. NON-TUNED, TWO-STAGE PIPELINE

Stage 1:

- Data: Commander, SMICA
- $\ell = 2-30$
- Statistics: alignment, entropy/tensor, phase correlation
- Control: Gaussian simulations

Stage 2:

- Map and release variation
- Fixed alternative  $\ell$ -range
- Monte Carlo repetition

Criteria:

- Positive: stable, multi-statistic signal
- Negative: Gaussian-compatible or unstable

## 8. ROLE OF THE TWO-STAGE STRUCTURE

Prevents post hoc tuning and ensures robustness.

## 9. POSITION OF THIS WORK

This is not a proof of PIUM, but a reproducible test framework.

## 10. FALSIFIABILITY

Rejected if consistent with Gaussian null hypothesis.

## 11. CONCLUSION

The low- $\ell$  CMB regime allows controlled testing of phase coherence. PIUM becomes testable, not yet confirmed.

## REFERENCES

Planck Collaboration (2020) V, VII  
Copi et al. (2010)  
Patel et al. (2025)  
Aluri et al. (2025)  
Billi et al. (2024)  
Owusu et al. (2023)

END